Quantum Field Theory – Homework 3

Name: ______________________ **Due Date:** 2025.11.10

The following problems are based on Lectures 5 and 6, focusing on the quantization of the harmonic oscillator and the free scalar field. The bullet points are guiding hints to help structure your answers.

Problem 1. Quantization of the Harmonic Oscillator (20 points)

Consider the one-dimensional harmonic oscillator with Lagrangian

$$L = \tfrac{1}{2}\dot{x}^2 - \tfrac{1}{2}\omega^2 x^2.$$

(a) Derive the conjugate momentum and Hamiltonian of the system.

(b) Quantize the system by promoting x and p to operators satisfying the canonical commutation relation $[\,\hat{x}, \hat{p}\,] = i$.

(c) Define the creation and annihilation operators

$$\hat{a} = \frac{1}{\sqrt{2\omega}}(\omega\hat{x} + i\hat{p}), \qquad \hat{a}^\dagger = \frac{1}{\sqrt{2\omega}}(\omega\hat{x} - i\hat{p}),$$

and derive their commutation relation.

(d) Express the Hamiltonian in terms of $\hat{a}$ and $\hat{a}^\dagger$, and find the general form of its energy eigenvalues and eigenstates.

Problem 2. Quantization of the Free Scalar Field (25 points)

Consider the free real scalar field with action

$$S = \int d^4x \left[\tfrac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) - \tfrac{1}{2} m^2 \phi^2 \right].$$

(a) Derive the field equation of motion and the conjugate momentum.

(b) Write down the general mode expansion of the quantum field operator $\hat{\phi}(x)$ in terms of creation and annihilation operators $\hat{a}_{\vec{k}}$ and $\hat{a}^\dagger_{\vec{k}}$.

(c) Starting from the canonical commutation relation $[\,\hat{\phi}(t,\vec{x}), \hat{\Pi}(t,\vec{x}')\,] = i\delta^{(3)}(\vec{x}-\vec{x}')$, derive the commutator between $\hat{a}_{\vec{k}}$ and $\hat{a}^\dagger_{\vec{k}'}$.

(d) Explain why each mode labeled by $\vec{k}$ behaves as an independent harmonic oscillator, and describe the physical meaning of $\hat{a}^\dagger_{\vec{k}}$ acting on the vacuum state.

Problem 3. Zero-Point Energy and Momentum Cutoff (20 points)

The vacuum energy density of the free scalar field is formally given by

$$\epsilon_0 = \frac{1}{2} \int \frac{d^3\vec{k}}{(2\pi)^3} \sqrt{\vec{k}^2 + m^2}.$$

(a) Introduce a momentum cutoff Λ and evaluate the leading dependence of ϵ_0 on Λ in the limit $\Lambda \gg m$.

(b) Explain the physical origin of the divergence in ϵ_0.

(c) Discuss briefly how this divergence relates to the continuum nature of field theory and why it leads to the so-called cosmological constant problem.